

where C is scaled for the length of the trading horizon and expressed in terms of $(\$/\text{share})^2$, c_{ij} is the covariance between stock i and stock j , and c_{ii} is the variance of stock i .

Market Impact

$$I_{\$i}^* = a_1 \cdot \left(\frac{X_i}{ADV_i} \right)^{a_2} \cdot \sigma_i^{a_3} \cdot 10^{-4} \cdot P_{i0} \cdot X_i \quad (9.1)$$

$$MI_{\$}(x_k) = \sum_{i=1}^m \left(\sum_{t=1}^n \frac{b_1 \cdot I_i^* \cdot x_{it}^2}{X_i \cdot v_{it}} \right) + (1 - b_1) I_i^* \quad (9.2)$$

This formulation of market impact follows from the trade schedule formulation with parameter $a_4 = 1$. These parameters are given in Table 9.1.

Price Appreciation

$$PA_{\$}(x_k) = \sum_{i=1}^n \sum_{t=1}^m x_{ij} \cdot \Delta p_i^* \cdot t \quad (9.3)$$

where Δp_i is the per period price appreciation term expressed in $\$/\text{share}$ adjusted for the side of the order. For example, if the price is expected to increase $\$0.05/\text{share}$ per period and we are selling shares the price appreciation term is:

$$\Delta p_i^* = \text{side}(i) \cdot \Delta p_i = -1 \cdot \$0.05/\text{share} = -\$0.05/\text{share}$$

Table 9.1 Market Impact Parameters—Trade Schedule Strategy

Data Sample	a_1	a_2	a_3	a_4	b_1
All Data	656	0.48	0.45	1	0.90
Large Cap	707	0.59	0.46	1	0.90
Small Cap	665	0.42	0.47	1	0.90